SecuredWave Cybersecurity Internship

Home Task — Wireshark Fundamentals Lab

Capture file: lab1-wiresharkintro.pcapng

Instructions: Open the provided capture file in Wireshark and answer the following questions in your own words. Include the exact display filter you used for each question where applicable. Submit your answers as a single document before the deadline given by your mentor.

Section A — Getting Oriented

1. Open the file and identify the three main panes of the Wireshark window. What does each one show?

2. How many packets are in this capture, and roughly how long did the capture last?

3. What IP address belongs to "our" machine (the one doing the browsing)?

Section B — Filtering Basics

4. Type ip.addr == 192.168.4.100 in the display filter bar. What happens, and why is this useful?

5. What's the difference between ip.addr == 192.168.4.100 and ip.src == 192.168.4.100?

6. Filter for tcp.port == 443. What do you see, and what does port 443 tell you about the traffic?

7. Clear your filter, then filter for just dns. How many packets match?

Section C — DNS

8. With the dns filter applied, list every hostname this machine looked up.

9. Pick one DNS response packet. What IP address did the server return for www.udemy.com?

Section D — HTTP

10. Filter for http. What do you find?

11. Select the GET request packet and expand it fully. What can you read in plain text that you could NOT read if this were HTTPS?

12. Look at the server's response to that GET request. What status code does it return, and what does that mean?

13. Right-click the HTTP GET packet → Follow → HTTP Stream. What's different about this view compared to the packet list?

Section E — TLS/HTTPS

14. Filter for tls.handshake.type == 1 (Client Hello). Open one and find the Server Name Indication (SNI). What does it tell you, given the packet is otherwise encrypted?

15. Filter for tls.handshake.type == 2 (Server Hello). What is the server telling the client in this packet?

16. Follow → TCP Stream on one of the port-443 conversations. What do you see, and why?

Section F — Statistics Menu

17. Open Statistics → Protocol Hierarchy. What does this tell you about the overall makeup of the capture?

18. Open Statistics → Conversations (IPv4 tab). Which remote host had the most packets exchanged with 192.168.4.100, and what does that suggest?

19. Open Statistics → Endpoints. How many unique IP addresses does 192.168.4.100 talk to in this whole capture?

Wrap-up Discussion

20. Walk through the full sequence of events in order: DNS lookup → TCP handshake → HTTP request → redirect → TLS handshake → encrypted data. Can you reconstruct this timeline using the packet numbers alone?

21. Why might a network admin still tell a user visited "udemy.com" even though the connection is HTTPS?

22. How would you actually decrypt the TLS traffic in Wireshark if you needed to?